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COMPUTER GRAPHICS

Section: L GROUP: 8

LAB REPORT ON:

Airport Simulation System

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1. Introduction

Computer graphics is widely used to create visual simulations and animations using programming. In this project an airport environment was designed using the OpenGL graphics library. Different objects such as airplanes, runway, buildings, trees, clouds, birds and towers were created to represent a realistic airport environment.

The purpose of this project was to demonstrate how different graphical primitives like polygons, triangles, lines and circles can be used to construct complex scenes. Some animations were also implemented in this project such as airplane takeoff, landing, bird movement and rainfall. Day and night transition was also implemented to make the scene more dynamic. The project was implemented using C++ and OpenGL with the GLUT library.

2. Proposal

The main objective of this project was to design and simulate a simple airport environment using computer graphics techniques. In this project several graphical objects were created such as runway, buildings, watch tower, control tower, trees, clouds and airplane. Animation techniques were used to simulate airplane movement. The airplane was designed in such a way that it could perform takeoff and landing operations.

Different environmental effects such as day and night cycle, rainfall and bird movement were also included to make the scene more interactive and realistic. The entire system was implemented using OpenGL primitives and transformation logic.

3. Schematic Diagram

The schematic layout of the project scene consists of several layers arranged from background to foreground. The sky area was drawn first, followed by the sun or moon. Then clouds and birds were rendered in the sky region. Airport buildings, towers, trees and grass were placed in the middle portion of the scene. Finally the runway and the moving airplane were drawn in the foreground.

4. List of Objects

- Airplane
- Runway
- Runway Lights
- Buildings
- Watch Tower
- Control Tower
- Trees
- Grass

- Sun and Moon
- Clouds
- Birds
- Rain System
- Stars

5. Functions to Represent the Objects

Function Name	Purpose
init()	Initializes OpenGL environment
drawRunway()	Draws airport runway
drawAirplane()	Creates airplane structure
drawBuilding()	Draws buildings
drawTree()	Draws trees
drawWatchTower()	Creates airport watch tower
drawControlTower()	Creates control tower
drawCloud()	Draws clouds
drawBird()	Draws flying birds
drawSunMoon()	Draws sun and moon
drawRain()	Creates rainfall effect
drawStars()	Draws night sky stars

6. Interactive Functions

Key	Function
L	Airplane Landing
T	Airplane Takeoff
R	Start Rain
S	Stop Rain
Q	Quit Program

7. Task Assignment and Codes of Functions

In this project multiple tasks were implemented to simulate a realistic airport environment. The OpenGL environment was initialized using the init() function. The projection system was configured so that a 2D coordinate system could be used.

The airplane structure was created using polygons and triangle fan primitives. Multiple vertices were used to design different parts of the airplane such as wings, body, tail and wheels. The runway was drawn using quadrilateral shapes and dashed lines were added to represent the center markings.

Buildings and towers were created using quadrilateral shapes. Trees were designed using rectangles and triangles to represent trunks and leaves. The sky background was created and during night time stars and the moon were displayed.

The airplane movement logic was implemented using position variables which were updated continuously to simulate motion. Landing and takeoff behaviors were controlled through keyboard interaction. A rainfall effect and bird movement were also implemented to make the scene more dynamic.

8. Conclusion

In this project an airport environment was successfully simulated using OpenGL graphics programming. Different graphical objects such as airplane, runway, towers, trees and buildings were created using primitive shapes.

Several animation and interactive features were implemented such as airplane takeoff, landing, bird movement, rainfall and day-night transition. This project helped in understanding how computer graphics concepts can be used to create interactive visual scenes and provided practical experience in OpenGL programming.